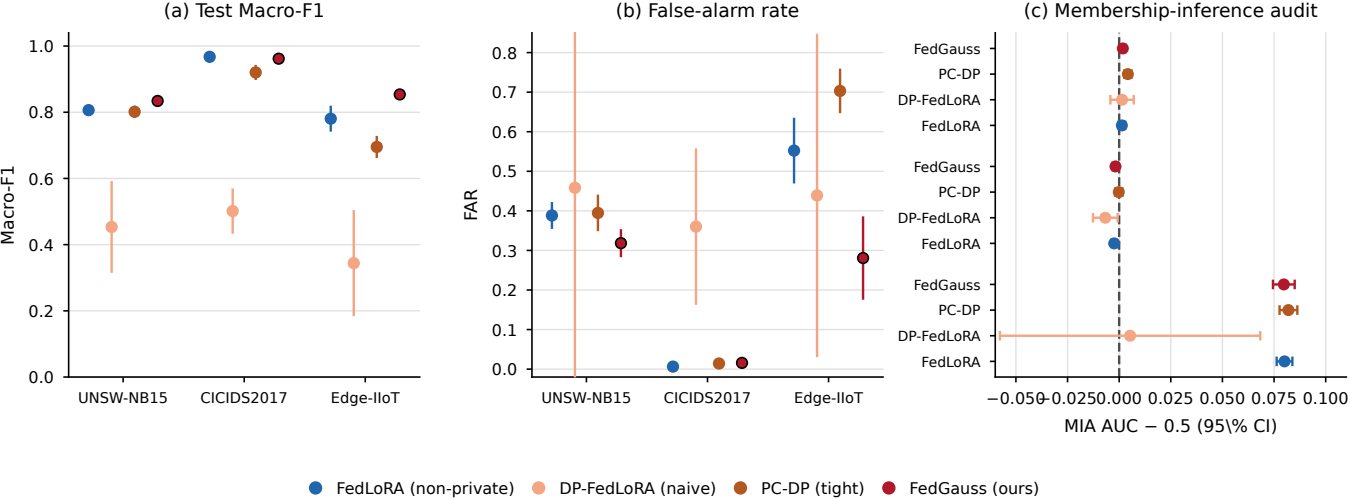
(a) Test Macro-F1
Macro-F1
1.0
0.8
0.6
0.4
0.2
0.0
UNSW-NB15
CICIDS2017
Edge-IIoT
(b) False-alarm rate
FAR
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
UNSW-NB15
CICIDS2017
Edge-IIoT
(c) Membership-inference audit
FedGauss
PC-DP
DP-FedLoRA
FedLoRA
FedGauss
PC-DP
DP-FedLoRA
FedLoRA
FedGauss
PC-DP
DP-FedLoRA
FedLoRA
−0.050
−0.025
0.000
0.025
0.050
0.075
0.100
MIA AUC − 0.5 (95\% CI)
FedLoRA (non-private)
DP-FedLoRA (naive)
PC-DP (tight)
FedGauss (ours)